

SONiX 32-Bit Cortex-M0 Micro-Controller

I2C_EEPROM Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/11/29	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of I2C EEPROM.

2 FEATURE

- This program provides an example for writing to and reading from an I2C EEPROM device which is device 0 (AO,A1 and A2 are all pulled to GND).
- Suit for different series of EEPROM.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Necessary to pull up I2C SCL and SDA pins with external pull-up resistors.
- Connect external EEPROM to starter kit board.

4 EXAMPLE CODE

1. Open I2C.h, and set EEPROM_less_than_32K to 0 or 1 on demand.

1.1 If EEPROM internally organized less than 4096*8

➤ #define EEPROM_less_than_32K 1

1.2 If EEPROM internally organized more than or equal to 4096*8

➤ #define EEPROM_less_than_32K 0

```

4  /* _____ INCLUDES _____ */
5  #include <stdint.h>
6
7  /* _____ DEFINITIONS _____ */
8  #define EEPROM_less_than_32K 1           // 1: for EEPROM size less than 32K bits (4096x8)
9                                           // 0: for EEPROM size more than or equal to 32K bits (4096x8)

```

2. Call SysTick_Init() and I2c_Init();

int main (void)

{

·
·

SysTick_Init(); //init SysTick

I2C_Init(); //init I2C0

·

·
·
User code
·
·

}

```

80  //System Interface Init
81  SysTick_Init();           //init SysTick
82  I2C_Init();               //init I2C0
83
84  while(1){
85
86  #if (EEPROM_less_than_32K == 0)
87      if(I2C_Write(0x00, 32))           //page write, data word address = 0x00 and write 32 bytes data to EEPROM
88  #else
89      if(I2C_Write(0x00, 8))           //page write, data word address = 0x00 and write 8 bytes data to EEPROM
90  #endif
91  {
92      UT_DelayNms (5);
93
94      #if (EEPROM_less_than_32K == 0)
95          if(I2C_Read(0x00, 32))           //read 32 bytes data from EEPROM
96      #else
97          if(I2C_Read(0x00, 8))           //read 8 bytes data from EEPROM
98      #endif
99      {

```

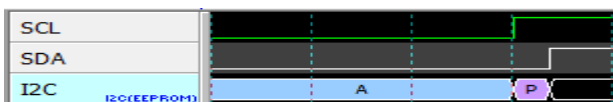
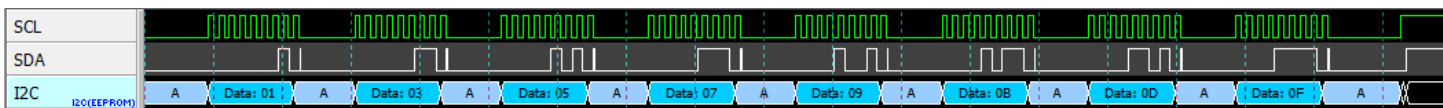
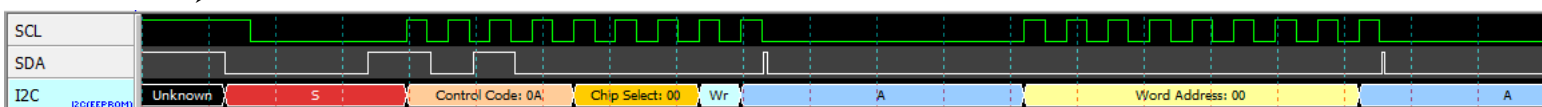
3. The program will be decided to write/read 32 bytes or 8 bytes of data in accordance with the EEPROM_less_than_32K EEPROM
4. If the data is correct, P2.1 will toggle.

5 RESULT

5.1 Write data

1. #define EEPROM_less_than_32K 1

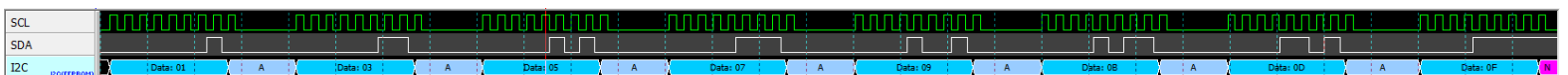
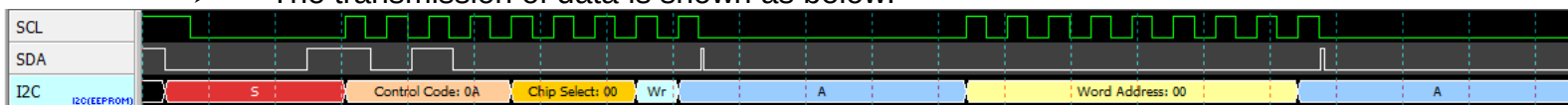
- Write 8 bytes data (1,3,5,7,9,B,D,F) to EEPROM
- The transmission of data is shown as below.
-



5.2 Read data

1. #define EEPROM_less_than_32K 1

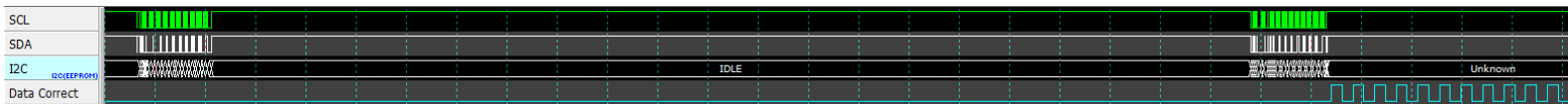
- Read 8 bytes data (1,3,5,7,9,B,D,F) from EEPROM
- The transmission of data is shown as below.



5.3 Compare data

1. If the data is correct, P2.1 will toggle.

- The transmission of data is shown as below.



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